Yeast Fermentation

Teacher guide | Graduated-cylinder lab (water displacement) | Biology 2e, Chapter 7

1. Organize a full comparison set

Question: How does the available sugar affect gas accumulation in a yeast culture? This download is specifically for graduated-cylinder lab (water displacement). Use the same method for all flasks in a pooled comparison.

Equipment and measurement

Matched flasks, fitted one-hole bungs and delivery tubing, water troughs, graduated cylinders, stands and clamps. Also use timers, thermometer, transfer tools, stocks and eye protection.

Gas enters an inverted, water-filled graduated cylinder and displaces water. The gas-water boundary provides a readable volume. This is water displacement, not collection by direct displacement of air.

One set = 9 independent flasks: N1-N3 (no added sugar), G1-G3 (glucose), S1-S3 (sucrose). Each flask is a fresh mixture. Multiple readings from one flask are not additional independent trials.

Students	Example allocation of the 9 flasks
18: nine pairs	Each pair prepares 1.
12: six pairs	Three pairs prepare 2; three pairs prepare 1.
10: five pairs	Four pairs prepare 2; one pair prepares 1.
9: three pairs and one trio	Each team prepares 2; the teacher prepares the ninth. Or assign the trio 3.

Assign one or two flask IDs per pair and record ownership on page 4. Roles: mix/prepare, measure/time, record/share. Pairs can combine roles. Each flask needs its own start time and elapsed-time record.

Demo: distribute observation and analysis roles while the teacher operates equipment. If using just one flask per treatment, report n = 1; do not call three time readings three trials. Share matched independent class trials to complete a set.

Allow 10-15 minutes to prepare, up to 30 minutes for gas-volume observations, stopping sooner at equipment capacity, and 15-20 minutes to analyze. Balloon work may need two lessons. The classroom pilot took longer than 30 minutes for balloon inflation; doubling the recipe is a new trial condition, not a guaranteed shorter lag.

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2. Stocks, flask sizes, and supply planning

Choose one flask size and recipe per pooled set. These are proposed pilot volumes, not a promise of measurable gas in one lesson. Keep all treatments at the same measured temperature; a teacher-managed 30 C water bath is one possible condition.

Flask capacity	Treatment stock*	Yeast suspension	Final liquid
50 mL	10 mL	10 mL	20 mL
100 mL	20 mL	20 mL	40 mL
Custom: ____ mL	____ mL	____ mL	____ mL

*Plain water for N; glucose stock for G; sucrose stock for S. Examples leave 60% headspace for gas/foam. Lower the fill if the pilot requires it; do not fill a 50 mL flask with 50 mL culture.

Revised classroom stocks - keep yeast and sugar separate

Batch option	Yeast stock	Each sugar stock
500 mL water per stock	20 g yeast + 500 mL water	80 g glucose OR sucrose + 500 mL water
1,000 mL water per stock	40 g yeast + 1,000 mL water	160 g glucose OR sucrose + 1,000 mL water

Target: 20 g yeast and 80 g sugar per 1,000 mL total water when combining matched batches. Prepare glucose and sucrose in separate labeled bottles. The no-added-sugar control uses plain water in place of sugar stock.

Volumes above mean WATER ADDED, not solution made up to final volume. Dissolved sugar increases volume. Equal measured stock aliquots are a practical approximation; combine equal fractions of the two prepared batches to preserve the exact per-water recipe. Do not label these as exact mass/volume percentages.

This is twice the earlier classroom recipe. It differs from the virtual 0.5 g yeast + 2 g sugar in 50 mL final mixture; do not compare absolute gas volumes as if recipes matched. Record glucose product and any recipe deviations.

Prepare labeled sugar stocks ahead. Prepare yeast close to use following package hydration guidance. Do not add sugar to shared yeast stock. Gently resuspend before EVERY aliquot because yeast settles. Use separate labeled transfer tools and equilibrate all stocks to the chosen test temperature.

For one 9-flask set	50 mL flasks	100 mL flasks
Plain water / glucose / sucrose (each)	30 mL	60 mL
Yeast suspension (all 9)	90 mL	180 mL
Sets from 1 L of EACH stock*	11	5

*Before losses; yeast is limiting. For B sets and V mL of each stock per flask: each treatment stock = 3 x B x V; yeast = 9 x B x V. Allow extra for spills, pilot and optional controls.

Chosen sugar stock: ____ g / ____ mL. Yeast: ____ g / ____ mL. Final sugar concentration: ____ %. Test temperature: ____ C.

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3. Set up, measure, and interpret

Fill the cylinder completely with water. Invert it with its mouth submerged in a trough, keeping out air bubbles, and clamp securely. Place the delivery-tube outlet under its submerged mouth. The flask end of the tube stays above the culture. Check for leaks and keep tubing open.

Label assigned flasks; dispense treatment stock first. Gently resuspend yeast stock immediately before each aliquot. Add yeast last, mix consistently, promptly connect the collector and start that flask's timer. Record the initial reading and any connection delay.

Read at 0, 5, 10, 15, 20, 25, 30 elapsed minutes, or a shared teacher-selected schedule. Read the collected gas volume at eye level from the cylinder scale at the gas-water boundary. Record the initial trapped-gas volume V0 and each raw volume Vt. Record net collected volume Vt minus V0. Keep the cylinder mouth submerged and the reading position consistent.

Record V0 before the first timed reading. Retain raw Vt readings; calculate net volume = Vt - V0. Flag an unexpected decrease rather than silently replacing it. M = missing; L = suspected leak. Record an observed zero as 0, but never replace missing data with zero.

Pooling and interpretation

Subtract each flask's own V0 before pooling. Average comparable net volumes and report actual n at each time. Keep raw readings and flag leaks, missing data and any exclusions; do not silently discard unusual results.

CO_2 can dissolve in the collection water, so collected volume may underestimate gas released. Unequal water levels create a pressure difference; keep the setup consistent and, when the teacher can safely do so, equalize inside/outside water levels for readings. Do not pool with syringe data.

Keep recipes, temperature, flask size, apparatus and units matched. Keep virtual and physical results separate even if both use mL. For an interval rate, divide the change in net collected volume by elapsed minutes (mL/min). A late interval rate and total gas accumulated are different quantities.

Do not require a winning sugar. Gas accumulation supports a claim about yeast activity; it does not measure ATP yield or establish oxygen absence. The no-added-sugar control can show background activity. An optional sugar-without-yeast control helps check physical expansion.

Practical checks

Wear eye protection; check glassware and follow school procedures. Do not taste cultures. Teacher manages warm water, foaming, cleanup and equipment capacity. Keep tubing unkinked and the cylinder mouth underwater. Stop before gas reaches the cylinder capacity. Disconnect the flask from the collection apparatus before cooling or dismantling to prevent water being drawn back into the culture.

If readings do not change: check yeast freshness, temperature, settling, leaks and collector performance. For balloons, inflation threshold may hide small gas amounts. Keep observations, allow more time if practical, and document problems.

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4. Assignment roster and sharing

Class/date: _____ Set ID: _____ Students: ____

Method: Graduated-cylinder lab (water displacement) | Unit: mL

Flask: ____ mL Treatment stock: ____ mL Yeast: ____ mL

Temperature: ____ C Reading interval: ____ min End time: ____ min

Flask ID	Treatment	Pair/team or demo	Start time
N1	No added sugar		
N2	No added sugar		
N3	No added sugar		
G1	Glucose		
G2	Glucose		
G3	Glucose		
S1	Sucrose		
S2	Sucrose		
S3	Sucrose		

Print worksheet pages 1-2 for each pair/team and pages 3-4 for each student or pooled group. Page 2 records up to two flasks; copy it for additional flasks. Page 3 combines all nine.

Report actual n if fewer than three valid independent trials remain. Keep flagged originals if a fresh replacement is made. Use the classroom worksheet or a separate spreadsheet. The virtual lab stays on a gas syringe; never enter balloon cm into its mL fields.

Method and reading connections

Science World: Yeast-Inflated Balloons https://www.scienceworld.ca/resource/yeast-inflated-balloons/ Science Buddies: An Aerobic Exercise (inverted-cylinder collection) https://www.sciencebuddies.org/science-fair-projects/project-ideas/MicroBio_p009/microbiology/yeast-metabolism-aerobic-anaerobic OpenStax Biology 2e, 7.5: Metabolism without Oxygen https://openstax.org/books/biology-2e/pages/7-5-metabolism-without-oxygen

The small-flask recipes and volumes are proposed pilot settings, not validated performance claims. The virtual apparatus measures collected gas; it does not reproduce every physical measurement error.